

AegisCare HMS / LabTrack v5.5 — DHA HIE (AfyaLink) FHIR R4 Exchange

System	AegisCare HMS / LabTrack v5.5
Standard	HL7 FHIR Release 4 (R4)
Target	Kenya DHA Health Information Exchange — Shared Health Record (AfyaLink)
DHA API base	https://ilm-dev.dha.go.ke/uat-middleware/api/v1
Document status	DRAFT for certification submission
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Section 1 — FHIR Version & Conformance

1.1 AegisCare generates FHIR R4 resources only. Resource JSON is produced by deterministic server-side builders (PL/pgSQL RPCs and Deno edge functions), not hand-written clinical documents.

1.2 Conformance targets (per DHA SHR documentation at <https://hie-docs.dha.go.ke/sharedhealthrecord/shr-records>):

- SHR accepts a FHIR Bundle of type collection with id (echoed as mediator_id); the middleware validates resourceType = Bundle.
- Accepted resource types: Encounter, EpisodeOfCare, Observation, Condition, Procedure, Medication, MedicationRequest, MedicationDispense, MedicationAdministration, Immunization, ServiceRequest, AllergyIntolerance.
- The Encounter must reference the visit's EpisodeOfCare; all clinical resources must reference that Encounter.
- Reads require the per-visit consent token (X-Consent-Token) and practitioner identifiers (X-PUID, practitioner_id).

1.3 Conformance status: ⚠ PARTIAL. Individual resources are structurally R4-conformant and intended for validation at <https://validator.fhir.org>, but full profile-level validation against the DHA Implementation Guide and end-to-end Bundle submission to the SHR sandbox have not yet been performed (blocked on credentials — DOC-7 §1, §10).

Section 2 — Supported FHIR Resources

Resource	Builder	Trigger / API	Status
Patient	supabase/functions/fhir-patient/index.ts (edge function)	POST body {patient_id}	live
Encounter	generate_fhir_encounter(p_encounter_id UUID) PL/pgSQL RPC (20260805000000_sprint52_sha_fhir_foundation.sql, extended 20260806000004_sprint8e_fhir_second_schedule.sql); supabase/functions/fhir-encounter/index.ts	RPC call or edge function	live (on-demand)
Condition[]	supabase/functions/fhir-condition/index.ts from encounter_diagnoses	POST body {encounter_id}	live (on-demand)

ResourceBuilder	Trigger / API	Status
trg_medication_dispense_fhir() trigger function MedicationDispense (20260814000001_sprint12d_medication_dispense_fhir.sql)	prescriptions.statuses	live
Bundle GAP — not yet built (collection)	required for POST /shr/bundles	pending (build + credentials)
EpisodeOfCare GAP — not yet built	required by SHR Encounter references recommended trigger on lab_results	pending
Observation GAP — not yet built (see §9)		pending

Section 3 — FHIR Encounter Resource

3.1 Database builder — generate_fhir_encounter(p_encounter_id UUID)
 (STABLE, SECURITY DEFINER, search_path = public, revoked from anon):

FHIR element	AegisCare mapping
id	encounters.id
identifier[0]	official — https://health.go.ke/facility/{KMHFL}/encounter-id
identifier[1]	secondary — https://sha.go.ke/ns/notification-number (encounters.sha_notification_number; TEMP-SHA-... placeholder until live)
status	waiting→arrived, done→finished, else in-progress
class	v3-ActCode: inpatient→IMP, emergency→EMER, else AMB
serviceType	http://dha.go.ke/fhir/CodeSystem/service-type — PHF/SHIF/ECCIF from encounters.sha_fund_type
subject	Patient/{id} + display name
participant	Practitioner/{created_by} + council registration identifier (https://hwr.health.go.ke/ns/council-registration) from profiles.council_*
period	created_at → updated_at
diagnosis[]	encounter_diagnoses ordered by sequence; role from diagnosis_type (billing/secondary); rank; ICD-11 URI extension
hospitalization	referral block (referral_direction, referral_out_facility, referral_out_reason)
serviceProvider	https://kmhfl.health.go.ke identifier + facility name from app_settings
extension[]	sha-notification-number, sha-fund-type, clinical-findings (vitals + history), blood-group, allergies (Second Schedule coverage)

3.2 HTTP builder — fhir-encounter edge function
 (supabase/functions/fhir-encounter/index.ts): same resource over HTTP with meta.profile, priority (emergency), diagnosis references to Condition/{id}, serviceProvider with https://hiskeny.org/facility identifier, and hospitalization.dischargeDisposition for referrals out.
 Content-Type application/fhir+json.

3.3 Usage: invoked by the insurance desk for claim payloads

(rooms.\$id.tsx submitClaim() → supabase.rpc("generate_fhir_encounter", ...) → stored in dha_outbound_queue.payload.fhir_encounter and previewed in admin.queue.tsx); sample output can be generated for the certification application with SELECT generate_fhir_encounter('<encounter-uuid>');

Section 4 — FHIR MedicationDispense Resource

4.1 Trigger function — trg_medication_dispense_fhir() (SECURITY DEFINER, search_path = public), attached as trigger trg_medication_dispense_fhir AFTER UPDATE on prescriptions (20260811000001_sprint12d_medication_dispense_fhir.sql).

4.2 Fires when prescriptions.status transitions to dispensed.

Builds:

FHIR element	Mapping
id	prescriptions.id
status	completed
medicationCodeableConcept	text = drug_name; coding system https://kems.go.ke/ns/nlmis-code when stock_item_id present
subject	Patient/{encounter.patient_id}
context	Encounter/{encounter_id}
performer	Practitioner/{dispensed_by} + dispensed_by_name
quantity	UCUM (http://unitsofmeasure.org, code U)
whenHandedOver	dispensed_at
dosageInstruction	dosage frequency duration text + patientInstruction (notes)
extension[]	encounter-type, https://kmhfl.health.go.ke/facility, facility-name

4.3 Outcome: the resource is inserted into dha_outbound_queue (queue_type = 'fhir_sync', status pending), and automatically mirrored to shr_transmission_log by trg_shr_transmission_log_insert — the statutory SHR transmission metadata log.

4.4 Current disposition: ☐ PENDING transmission — the queue is live and monitored in admin.queue.tsx; actual POST /shr/bundles delivery is blocked on AfyaLink credentials (DOC-7 §10 item 6).

Section 5 — FHIR Claim Bundle — Current Status

5.1 Built (live in the database):

- sha_claims.fhir_bundle column (JSONB) — populated by the insurance desk with the generate_fhir_encounter() payload at submission time.
- sha_claim_items with intervention_code (SHA intervention mapping field).
- sha_benefit_packages codes (e.g., SHA-19-SC-13-style sub-benefit codes supported via intervention_code).
- Claim status machine on sha_claims (draft → pending otp → pending_preauth → ready → submitted → acknowledged → approved → rejected → appealed → paid).
- claims-dispatcher ShaClaimsHandler — stub that queues queue_type = 'sha_claim' with the FHIR payload.

5.2 Pending — "SHA-10" (SHA claims bundle / FHIR Claim resource):

-  GAP (build): no FHIR R4 Claim/ClaimResponse resource builder exists yet (the DHA/SHA eClaims pipeline expects claim submission with FHIR-encoded clinical content; today AegisCare attaches the Encounter resource inside the queue payload rather than a full claim Bundle).
-  GAP (build): deterministic claim Bundle assembly (Patient + Encounter + Condition[] + MedicationDispense[] + Claim) is the recommended next deliverable, aligned to the DHA eClaims catalogue (<https://hie-docs.dha.go.ke/eclaims>).
-  PENDING (credentials): live claim submission and status polling (dha_claim_id assignment, claim_status transitions from SHA).

5.3 Workaround today: claims are fully tracked locally (claim number, status, submitted-at) and queued with FHIR evidence; the SHA portal/API submission is the remaining leg, documented step-by-step in [docs/integration-activation-manual.md](#) (Integration 4).

Section 6 — DHA Outbound Queue Architecture

6.1 Table dha_outbound_queue (20260812000006_drift_dha_outbound_queue.sql):

Column	Purpose
queue_type	fhir_sync
payload	JSONB — the FHIR resource(s) or claim payload
status	pending → processing → sent → acknowledged
attempts, last_attempted_at	retry bookkeeping
response, error_message	integration responses (e.g., mediator_id)

6.2 Producers:

- trg_medication_dispense_fhir() → fhir_sync (MedicationDispense)
- trg_shr_access_notification() → shr_access_notification (on every new encounter; also on break-glass access)
- claims-dispatcher edge function → fhir_sync / sha_claim / private_claim / cash_receipt (consent-gated for FHIR sync)
- Insurance desk submitClaim() → sha_claim / private_claim directly

6.3 Consumers: the dispatcher's handlers (stub mode today) and the admin queue monitor (admin.queue.tsx). When credentials arrive, handlers switch to live HTTP calls (OAuth2 token → API call → response/status update → retry on failure with attempts increment).

6.4 Observability: every fhir_sync / shr_access_notification transition is mirrored to shr_transmission_log (transmission metadata, response code, error, payload keys) — satisfying the Digital Health (Data Exchange Component) Regulations 2025 transmission-log obligation.

Section 7 — ICD-11 Integration

7.1 Structured coding (primary path):

- encounter_diagnoses — icd11_code, icd11_title, icd11_uri, diagnosis_type (incl. final/working/admission/discharge), sequence (rank 1 = primary, SHA-compliant ordering).

- `trg_sync_encounter_diagnoses` keeps the structured table in sync with `encounters.diagnoses JSONB` (20260730120000_dha_icd11_encounter_diagnoses_sync.sql).
- `clean_and_validate_diagnosis_insert()` validates/normalises codes via `validate_and_get_icd11()` against the local `icd11_codes` reference table.

7.2 WHO API (live search path):

- `supabase/functions/icd11-search/index.ts` — proxies to `https://id.who.int/icd/release/11/2024-01/mms/search` (flexisearch, API-Version: v2), OAuth2 client_credentials against `https://icdaccessmanagement.who.int/connect/token` with in-memory token caching; returns `{code, title, uri}`.
- ☐ PENDING: requires secrets `ICD_CLIENT_ID` / `ICD_CLIENT_SECRET`; until configured, the UI (consultation ICD-11 search in `rooms.$id.tsx`) falls back automatically to the local `icd11_codes` table — no code change needed (Task F, docs/dha-compliance-task-a-icd11-sync.md).

7.3 FHIR mapping: `Condition.code.coding.system = icd11_uri` (default `http://id.who.int/icd/release/11/mms`); Encounter diagnosis extension `https://health.go.ke/fhir/StructureDefinition/icd11-uri`.

Section 8 — NLMIS Medication Coding

8.1 `stock_items.nlmis_code` (20260810000001_sprint11c_nlmis_code.sql) — KEMSA/NLMIS commodity codes attached to 297 of 331 stock items (89.7%). The remaining 34 items are legitimately untagged (ARVs, EPI vaccines, equipment — not NLMIS-tracked commodities). `stock_store_balances_view` exposes `nlmis_code` for supply-chain reporting.

8.2 FHIR linkage: `MedicationDispense.medicationCodeableConcept.coding.system = 'https://kemsa.go.ke/ns/nlmis-code'` is emitted when `prescriptions.stock_item_id` is set (dispense-time), giving the SHR/NLMIS a machine-readable commodity code.

8.3 Status: ☐ coding layer COMPLIANT. GAP (build): NLMIS/KHIS reporting submission integration is not built (MOH 707 internal reporting exists); see DOC-7 §8.

Section 9 — Pending Integrations

Item	Type	Blocked by
FHIR Bundle builder + POST <code>/shr/bundles</code>	build gap + credentials	DHA OAuth2 + AfyaLink sandbox (Provisional Interoperability Accreditation)
EpisodeOfCare generation	build gap	—
FHIR Observation/DiagnosticReport for lab results	build gap	—
FHIR Claim bundle (SHA-10)	build gap	SHA claims API credentials
Live SHR consent flow (POST <code>/shr/consents</code> , verify, X-Consent-Token)	credentials	DHA consent services access
SHR reads (GET <code>/shr/patient-records</code> , GET <code>/shr/Observation</code>)	credentials	consent-token flow live
HWR practitioner identifiers (PUID)	credentials	Kenya HWR API key
WHO ICD-11 API keys	credentials	<code>ICD_CLIENT_ID</code> / <code>ICD_CLIENT_SECRET</code>

Recommended activation sequence is documented in
docs/integration-activation-manual.md (Integrations 1–6) and DOC-7 §10.

End of DOC-5.